

Molecular Basis of Polymer mechanics

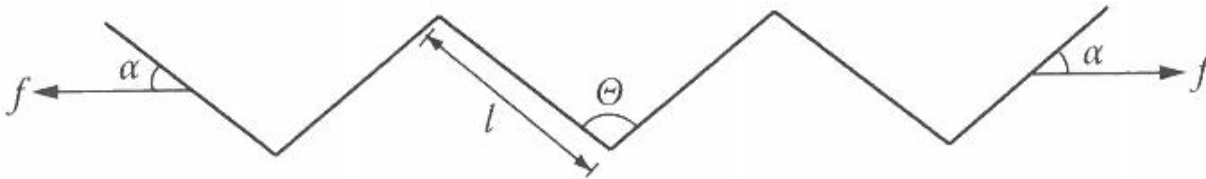
- i) Molecule-level basis of elastic deformation
- ii) Molecule-level understanding of stress relaxation
- iii) Other models of mechanical behavior

Modeling Mechanical properties of Polymers

- i. Molecular basis (Treolar's model)
- ii. Macroscale models (spring/Dashpot)

Elastic deformation of a Polymer chain

- From crystal structure & force constants of chemical bond, we can calculate a theoretical moduli of polymer crystals from deformation of single polymer chains.
- Assume all trans-polymer chain conformation



- Assume polymer chain is made up of n rods, length, l , joined by torsional springs
- Rods stretch along the length and no bending.
- Original length of the chain,

$$L = nl \cos \alpha$$

hence δL , upon deformation, is;

$$\delta L = n \delta(l \cos \alpha) = n(\cos \alpha \delta l - l \sin \alpha \delta \alpha) \quad \dots\dots(i)$$

- Component of force acting along bond direction is;

$$\delta l = \frac{f \cos \alpha}{k_l} \quad \dots\dots(ii)$$

where K_l = bond force constant

Elastic deformation of a Polymer chain

- Force needed to cause bond-angle distortions relates to torque acting on each bond

$$\text{torque} = 0.5fl \sin \alpha$$

Hence;

$$\delta\Theta = \frac{fl \sin \alpha}{2k_{\Theta}} \text{ where } K_{\Theta} = \text{angular deformation force constant}$$

- But since $\alpha = (90^\circ - 0.5\theta)$, then $\delta\alpha = -0.5\delta\Theta$

Therefore;

$$\delta\alpha = -\frac{fl \sin \alpha}{4k_{\Theta}} \dots\dots\dots(\text{ii})$$

- Combine eq. i, ii, iii we get;

$$\frac{\delta L}{f} = n \left[\frac{\cos^2 \alpha}{k_l} + \frac{l^2 \sin^2 \alpha}{4k_{\Theta}} \right]$$

- The Young's modulus of the polymer is therefore;

$$E = \frac{(f/A)}{(\delta L/L)} \text{ (i.e. stress/ strain)}$$

Elastic deformation of a Polymer chain

- But A is the cross-section area supported by a polymer chain, therefore;

$$E = \frac{l \cos \alpha}{A} \left[\frac{\cos^2 \alpha}{k_l} + \frac{l^2 \sin^2 \alpha}{4k_\Theta} \right]^{-1}$$

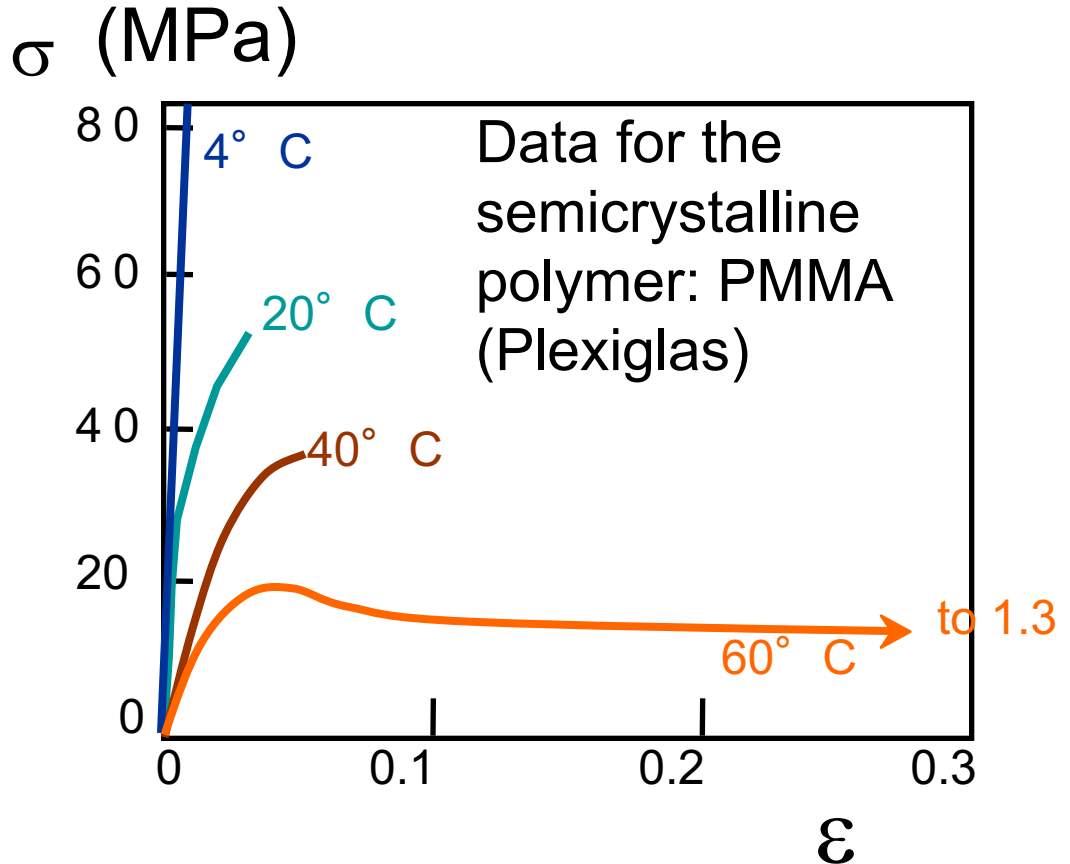
- Similarly, in terms of the bond angle;

$$E = \frac{l \sin(0.5\Theta)}{A} \left[\frac{\sin^2(0.5\Theta)}{k_l} + \frac{l^2 \cos^2(0.5\Theta)}{4k_\Theta} \right]^{-1}$$

- Therefore, we can estimate E if we know the crystal structure (A , l and Θ) and from IR or Raman spectra of a sample (k_l and k_Θ).
- NB: this method was initially developed by Treloar and is sometimes be referred to as “**Treloar’s approach**”*
- Weakness of this model:
 - Does not account for inter- and/or intra-molecular interactions
 - Gives the chain direction modulus – doesn’t fully explain elastic behavior of polymer
 - A complete set of elastic constants are needed

T and Strain Rate: Thermoplastics

- Decreasing T ...
 - increases E
 - decreases % EL
- Increasing strain rate...
 - same effects as decreasing T .



Objectives:

By the end of the lesson, we should:

- i) Explain stress-strain responses of different polymeric materials including block polymers (solids)
- ii) Be able to demonstrate why chain or fiber alignment induces anisotropy in mechanical behavior of a material
- iii) Estimate the chain-direction Young's modulus of a polymer based on its crystal structure and IR/Raman spectra
- iv) Extend our approaches to modelling complex polymer systems
- v) Understand how E changes with T over multiple polymer classes
- vi) Relate elasticity to thermodynamics

Modeling Mechanical properties of Polymers

- i. ~~Molecular basis (Treolar's model)~~
- ii. Macroscale models (spring/Dashpot)

Viscoelasticity

- Mechanical behavior of polymers to applied stress or strain depends on or of loading
- – at low strain-rate, obey *Hooke's law*
i.e. $\sigma = E\varepsilon$
- – at low strain-rate, obey *Newton's law*
i.e. $\sigma = \eta d\varepsilon/dt$
- Most polymers are in between viscous liquids and elastic materials (viscoelastic)
- Viscoelasticity is highly dependent on temperature.

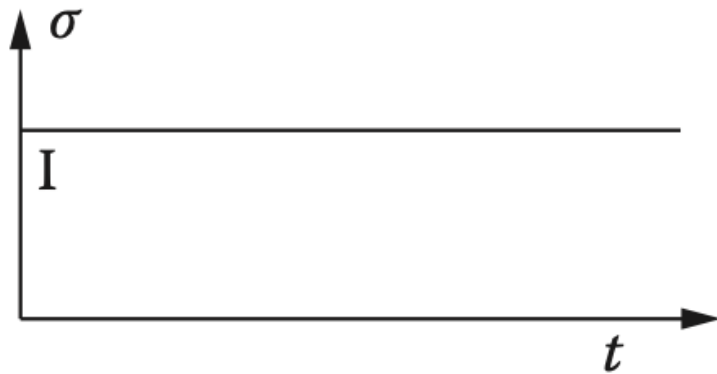
why (molecular level)?

Viscoelasticity

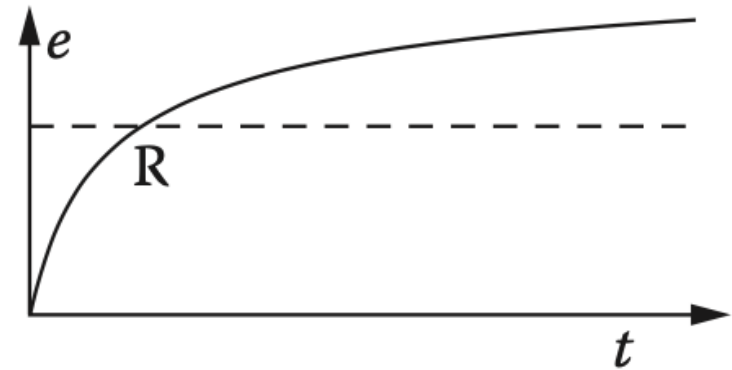
- What happens when weight is suspended on a polymer filament?
 - Strain increases over time due to stress-induced molecular rearrangement in the solid. Upon release, the molecules slowly recover to original state
 - This is called – a manifestation of viscoelasticity.
- How then can we use polymers in engineering?
 - Polymers do not creep and they fully recover on removing the stress
- Thus is important in the use of polymer as an engineering material.

Viscoelastic responses: creep

Stress system

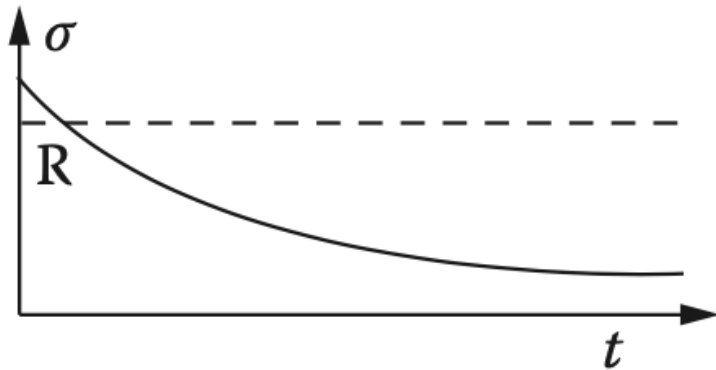


Strain system

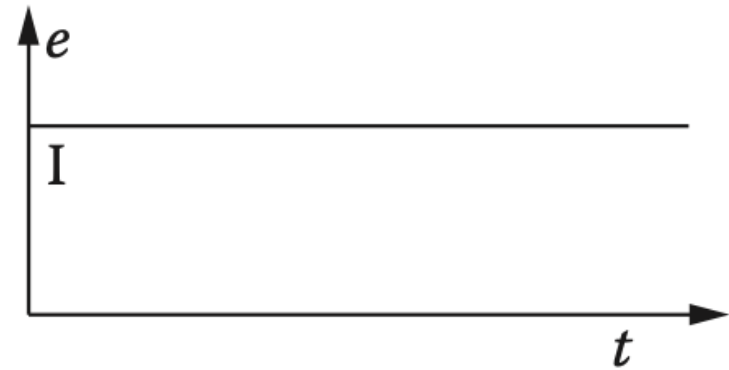


Viscoelastic responses: relaxation

Stress system

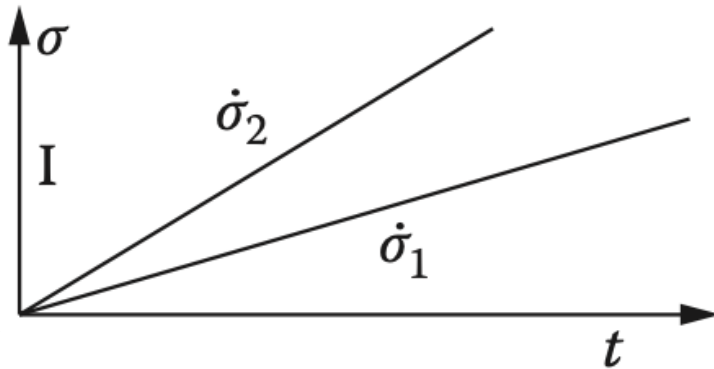


Strain system

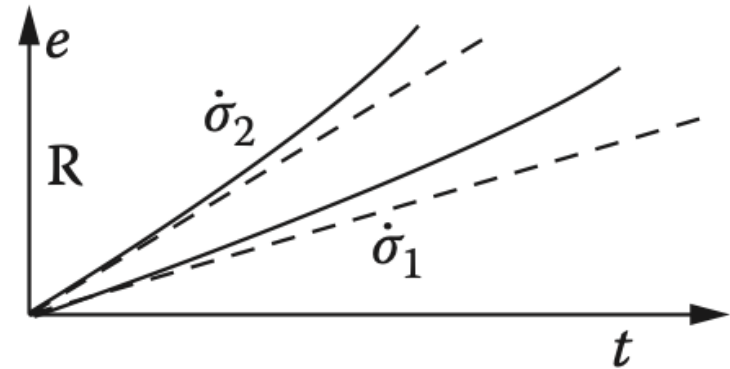


Viscoelastic responses: constant stress-rate

Stress system

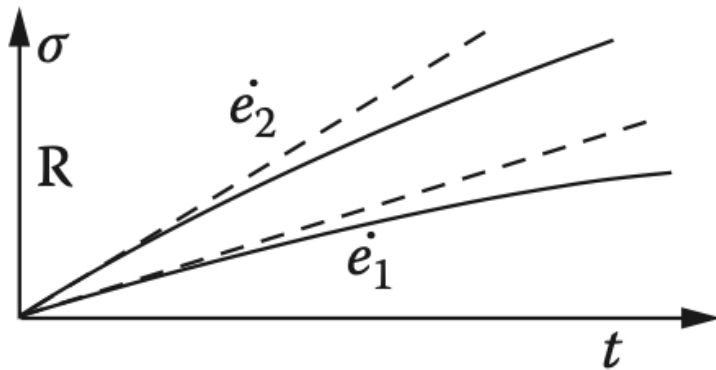


Strain system

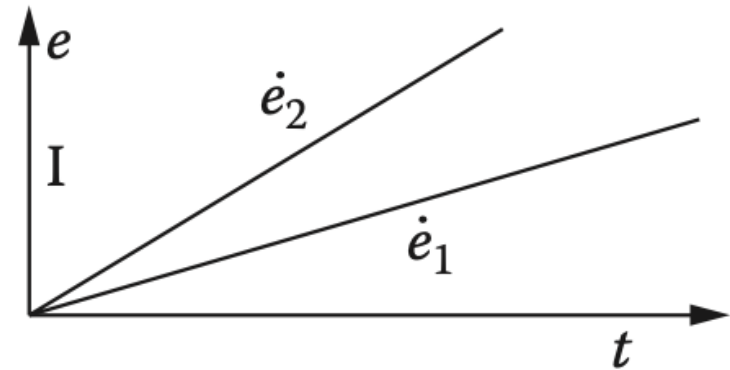


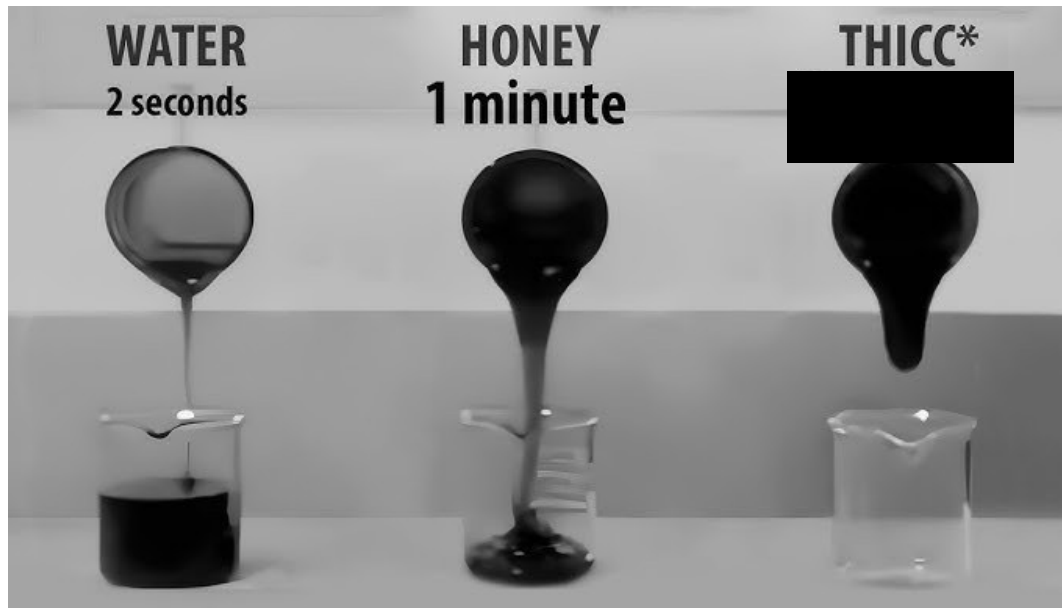
Viscoelastic responses: constant stress-rate

Stress system



Strain system



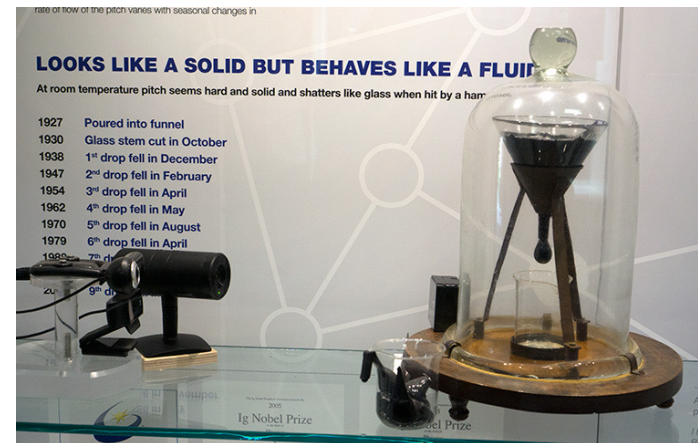


How long does it take for 1 drop of
pitch to form & release?

GROUP ACTIVITY!

Extremes of viscoelasticity: pitch

- Pitch: extremely viscous (polymer)
- Produced through distillation of carbonaceous materials
- Rosin: derive from plants (usually pine trees)
- Bitumen/asphalt: derive from petroleum

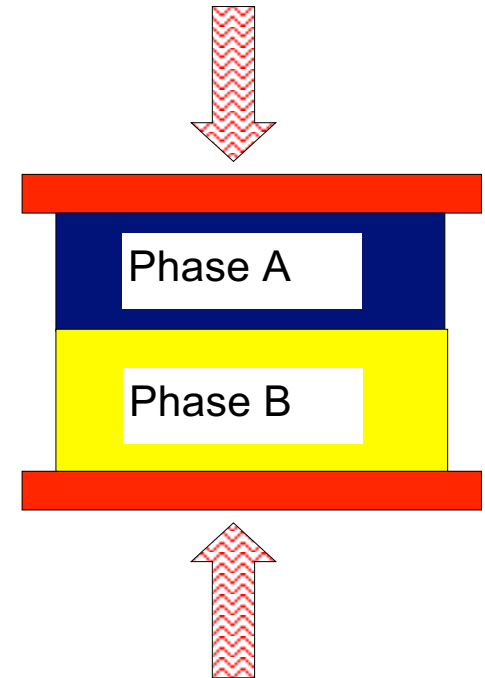


https://en.wikipedia.org/wiki/Pitch_drop_experiment

<https://vimeo.com/event/4721898>

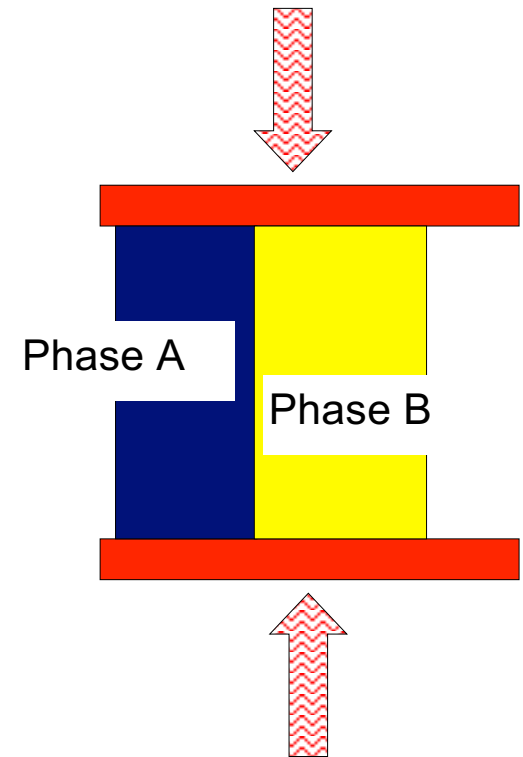
Isostress

- Imagine parallel slabs of material between platens that apply a load.
- Phase A has volume fraction V_A and modulus E_A ; Phase B has volume fraction V_B and modulus E_B .
- What is the composite modulus, E_C ?
- We assume because each phase sees the same stress (assuming same cross-sectional area).
- The stress, $\sigma = \sigma_C$, is therefore the field and the strain is the response (and the compliance is the property).



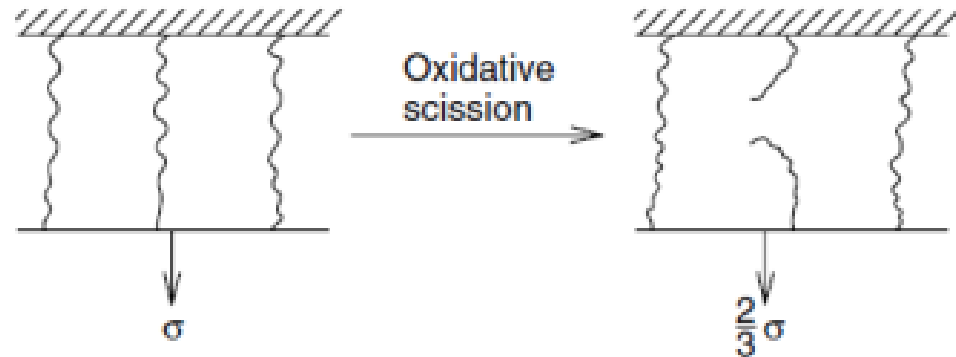
Isostrain

- Imagine parallel slabs of material between platens that apply a load.
- Phase A has volume fraction V_A and modulus E_A ; Phase B has volume fraction V_B and modulus E_B .
- What is the composite modulus, E_C ?
- We assume because each phase sees the same change in length.
- The strain, $\varepsilon = \varepsilon_c$, is therefore the field and the stress is the response (and the stiffness is the property).

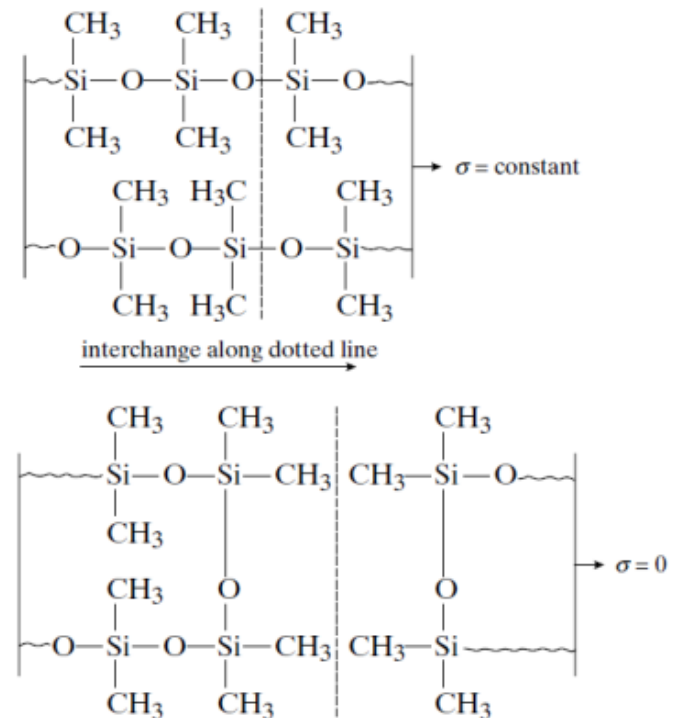


Molecular basis stress relaxation

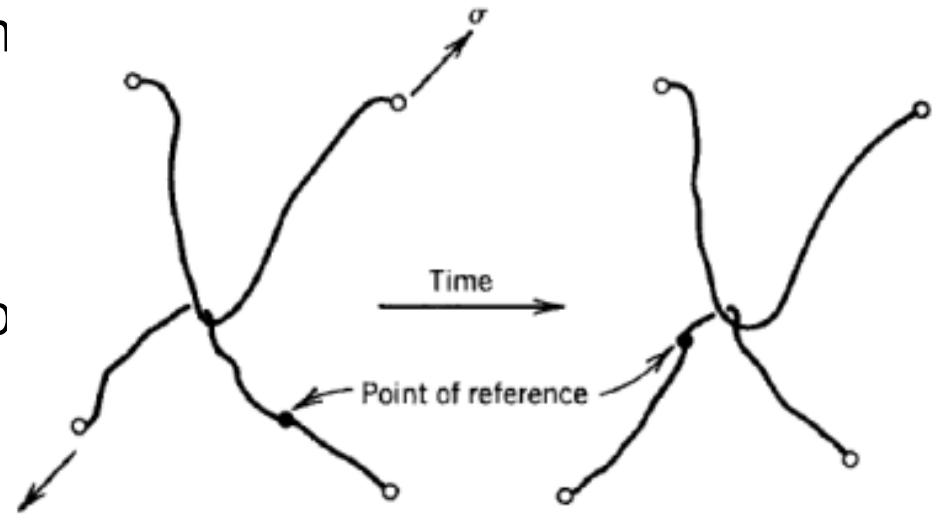
- : breakage of polymeric chains – reduces mechanical strength but increases the elongation during creep



- : chains exchange bonds leading to chain re-organization
 - Common in polysiloxanes and polyesters



- **Molecular relaxation**: energy dissipation through molecular vibrations (energy dissipation)
- **Thirion relaxation**: entangled chains slip *reversibly* over each other
 - Specific to polymers
- **Viscous flow**: polymer chains slip past each other under flow.
 - Permanent
 - More prominent in polymers



But how can we model/understand relaxation and creep?

- Two approaches: stress relaxation or strain relaxation

Understanding stress relaxation & creep

Maxwell model (isostress)

- We employ and
- Dashpot: a mechanical device that resists motion through viscous friction e.g. used in hydraulic doors
 - The mechanical behavior is described as;

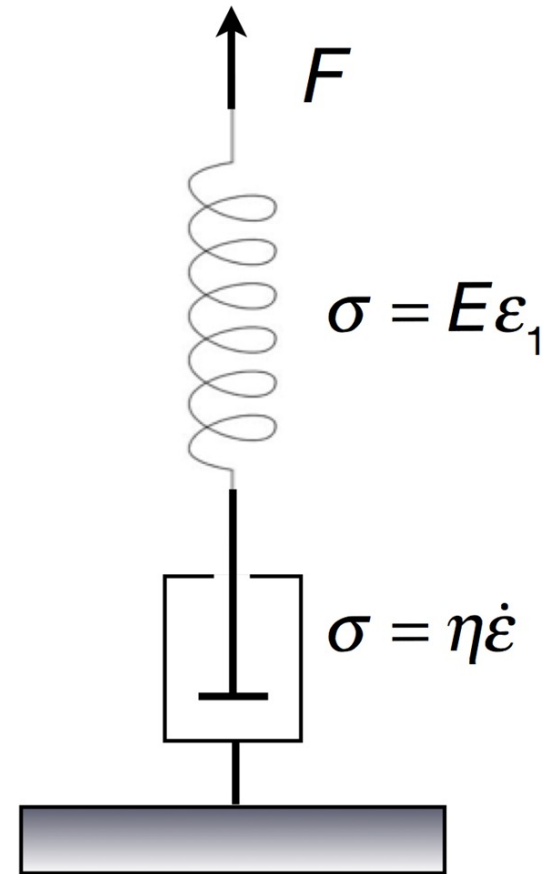
$$\sigma = \eta \frac{d\varepsilon}{dt} = \eta \dot{\varepsilon}$$

- Spring: Elastic behavior follows Hooke's law, response to stress is;

$$\sigma = E\varepsilon$$

- Maxwell model places a spring and dashpot in series.
 is same across the assembly, but is different, hence;

$$\frac{d\varepsilon}{dt} = \frac{d\varepsilon}{dt} + \frac{d\varepsilon_1}{dt} = \frac{1}{E} \frac{d\sigma}{dt} + \frac{\sigma}{\eta}$$



Understanding stress relaxation & creep

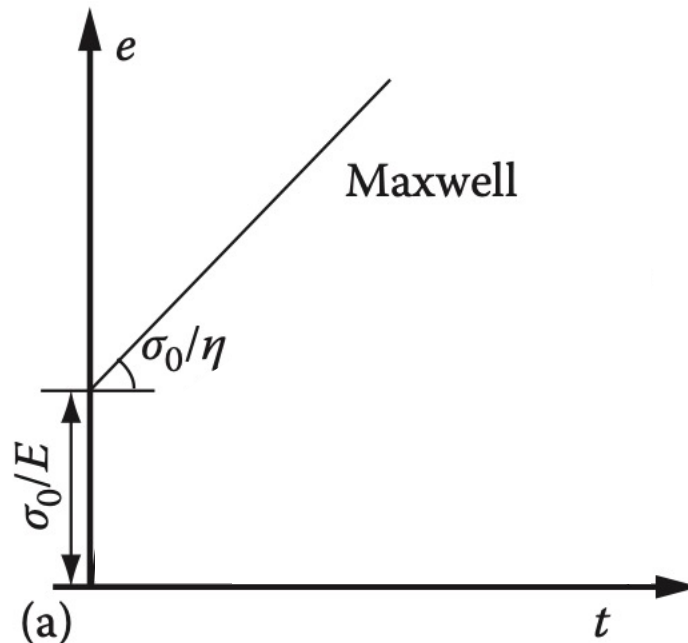
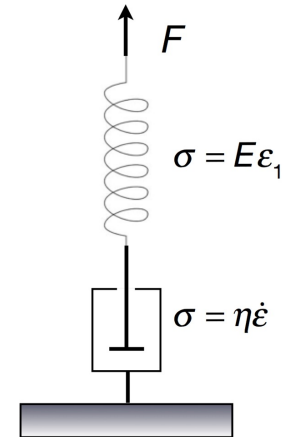
Maxwell model

- Creep = constant stress:

$$\sigma = \sigma_0$$
$$\frac{d\sigma}{dt} = 0$$

Therefore

$$\frac{de}{dt} = \frac{\sigma_0}{\eta}$$



Understanding stress relaxation & creep

Maxwell model

- Stress relaxation = constant strain

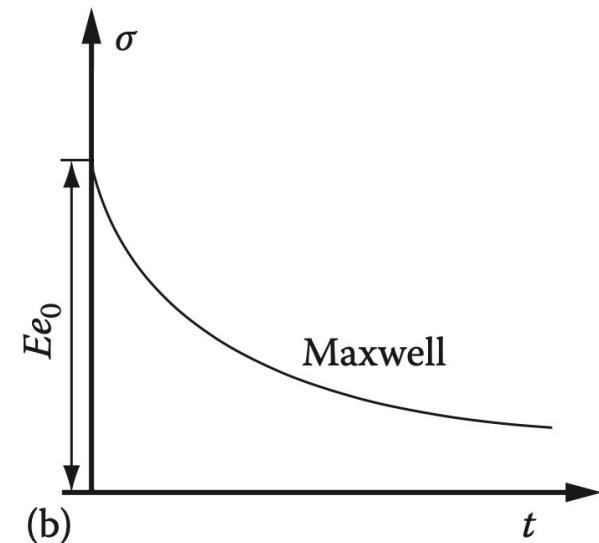
$$e = e_0$$

$$\frac{de}{dt} = \frac{1}{E} \frac{d\sigma}{dt} + \frac{\sigma}{\eta} = 0$$

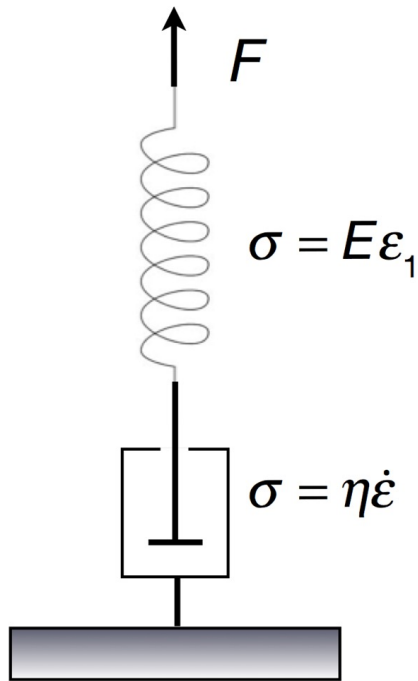
- Then;

$$\frac{1}{E} \frac{d\sigma}{dt} = -\frac{\sigma}{\eta} \Rightarrow \frac{d\sigma}{\sigma} = -\frac{E}{\eta} dt$$

$$\sigma = \sigma_0 e^{-(E/\eta)t}$$



Stress relaxation & creep: Maxwell model



- Can we estimate the relaxation time?
- Will the stress eventually dissipate and system relaxes back?
- Relaxation time, τ , is defined by;

$$\tau = \frac{\eta}{E}$$

- Hence;

$$\sigma = \sigma_0 e^{-(E/\eta)t}$$

becomes

$$\sigma = \sigma_0 e^{-t/\tau}$$

